

UNIT-1 Design Thinking Introduction

Definition/Concept

Design Thinking is a human-centered problem-solving approach that combines creativity, empathy, and logic to create solutions that truly meet people's needs.

It's widely used in business, education, healthcare, and social innovation.

Why It Matters:

- Solves complex problems in innovative ways.
- Improves products, services, and experiences.
- Builds skills in creativity, collaboration, and adaptability.

Key Characteristics:

Human-Centered – Focuses on understanding the people you're designing for.

Creative – Encourages out-of-the-box ideas.

Iterative – Involves trying, testing, and improving solutions.

Collaborative – Uses diverse perspectives for richer results.

Why Design Thinking is Important

- Puts People First
- Ensures solutions are based on real human needs, not just assumptions.
- Increases user satisfaction and acceptance.
- Encourages Innovation
- Promotes creative, out-of-the-box thinking.
- Helps discover fresh solutions for old or complex problems.
- Solves Complex Problems
- Useful for “wicked problems” with no clear answers.
- Breaks challenges into manageable, testable steps.
- Reduces Risk of Failure
- Early prototypes and testing prevent costly mistakes.
- Feedback-driven improvements make solutions more effective.
- Works Across Industries
- Applicable in business, education, healthcare, government, and social projects.
- Useful for products, services, processes, and experiences.
- Builds Essential Skills
- Boosts creativity, empathy, collaboration, and adaptability.
- Prepares individuals and teams for real-world challenges.

History of Design Thinking

Period / Year	Key People / Organization	Milestone in Design Thinking History
1950s–1960s	John E. Arnold (Stanford), Herbert A. Simon	Early ideas of <i>creative problem-solving</i> and <i>design as a science</i> emerge. Herbert Simon’s book <i>The Sciences of the Artificial</i> (1969) lays the foundation for a structured design process.
1970s	Robert McKim (Stanford)	Introduces <i>visual thinking</i> and <i>integrated problem-solving</i> approaches in engineering design education.
1970	Horst Rittel & Melvin Webber	Introduced the concept of “wicked problems” – complex, ill-defined problems without clear solutions (e.g., poverty, urban planning). They argued that such problems needed collaborative, iterative, and user-focused methods.
1980s	Rolf Faste (Stanford)	Expands McKim’s work, formalizes “design thinking” as a way to apply designers’ methods to problem-solving beyond product design.
1991	David Kelley & IDEO	IDEO was formed (merger of David Kelley Design, ID Two, Matrix Product Design) and popularized <i>human-centered design</i> in business and product development.
2005	Stanford d.school (Hasso Plattner Institute of Design)	Launches academic programs for teaching <i>Design Thinking</i> as a 5-stage process: Empathize, Define, Ideate, Prototype, Test.
2009 onwards	Tim Brown (CEO of IDEO)	Published <i>Change by Design</i> , bringing design thinking into mainstream business strategy.
2010s–Present	Global businesses, NGOs, governments	Design thinking is widely adopted in sectors like healthcare, education, IT, government policy, and social innovation.

Conventional Problem-Solving vs. Design Thinking

Aspect	Conventional Problem-Solving	Design Thinking
Approach	Linear and analytical	Iterative and creative
Focus	Problem-focused (fix what's broken)	Human-focused (understand people's needs first)
Process	Define → Analyze → Solve	Empathize → Define → Ideate → Prototype → Test
Basis	Logic, past data, expert opinion	Empathy, experimentation, user feedback
Outcome	One "best" solution chosen early	Multiple ideas tested, refined, and evolved
Risk Handling	Avoids failure by planning extensively	Accepts failure as learning through quick tests
Innovation Level	Often incremental	Often innovative and disruptive
Collaboration	Can be siloed by department/expertise	Multidisciplinary teamwork
Timeframe	Spends more time upfront on analysis	Moves quickly from idea to prototype

In short:

- **Conventional problem-solving** = Fixing a problem using analysis and logic, often in a straight line.
- **Design Thinking** = Exploring a problem creatively with empathy, prototyping, and iteration until a user-centered solution emerges.

Examples of business uses of Design Thinking

Industry	Example	How Design Thinking Was Used	Outcome
Technology	IBM	Applied design thinking workshops to improve internal collaboration and create user-focused software interfaces.	Reduced product development cycle time by 33%.
Retail	Nike	Used empathy research to design <i>Nike Flyknit</i> , a lightweight, eco-friendly	Boosted sales and sustainability brand image.

Industry	Example	How Design Thinking Was Used	Outcome
		shoe tailored to athletes' needs.	
Hospitality	Airbnb	Conducted deep user interviews with hosts and guests to redesign the booking experience and trust-building features.	Transformed from near-bankruptcy to a multi-billion-dollar company.
Healthcare	Mayo Clinic	Used observation in the empathy stage to redesign patient waiting areas and care flow.	Improved patient satisfaction and reduced wait times.
FMCG	Procter & Gamble (P&G)	Applied design thinking to develop <i>Swiffer</i> after observing cleaning habits in homes.	Created a billion-dollar product line.
Automotive	Ford	Used rapid prototyping and user testing to create SYNC in-car connectivity systems.	Enhanced customer experience and brand loyalty.
Banking & Finance	Bank of America	Developed the "Keep the Change" program after studying customer saving habits.	Millions of new accounts and increased savings engagement.

TECHNICAL APPLICATIONS OF DESIGN THINKING



SOFTWARE DEVELOPMENT & IT

Redesigning a hospital's patient management software

• Empathy • Define • Ideate • Test



ENGINEERING & PRODUCT DESIGN

Developing an eco-friendly water purification device for rural india

• Understand • Prototype • Test



ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Building an AI chatbot for banking services

• Empathy • Define • Prototype • Test



MECHANICAL & ELECTRICAL ENGINEERING

Redesigning EV charging stations

• Empathy • Ideate • Prototype • Test



HEALTHCARE TECHNOLOGY

Portable infant warmers (like the Embrace Warmer)

• Identify • Prototype • Test



AEROSPACE & DEFENSE

NASA applied design thinking in planning

Business Uses of Design Thinking

1. Product Innovation

Purpose: Create new products that truly meet customer needs.

Example: Apple used design thinking to develop the iPhone by focusing on user experience rather than just technical specs.

2. Service Design & Improvement

Purpose: Enhance customer service, reduce friction in processes.

Example: Airbnb redesigned its booking process by interviewing hosts and guests, improving trust and ease of use.

3. Business Model Innovation

Purpose: Rethink how a company creates, delivers, and captures value.

Example: Netflix moved from DVD rentals to streaming after understanding shifting customer behaviours and frustrations.

4. Customer Experience (CX) Enhancement

Purpose: Make every customer interaction more engaging and satisfying.

Example: Starbucks applied design thinking to create a more personalized in-store and digital ordering experience.

5. Organizational Change & Culture Building

Purpose: Improve internal collaboration, employee engagement, and problem-solving.

Example: IBM trained employees worldwide in Enterprise Design Thinking, fostering cross-team innovation.

6. Process Optimization

Purpose: Make internal workflows faster, cheaper, and more user-friendly for employees.

Example: Procter & Gamble redesigned supply chain processes to be more responsive to customer demand.

7. Social Impact & Sustainability

Purpose: Develop solutions for social, environmental, and community issues.

Example: Unilever uses design thinking to develop sustainable packaging solutions that reduce waste.

8. Digital Transformation

Purpose: Guide technology adoption based on real user needs.

Example: Bank of America used design thinking to create “Keep the Change,” a digital savings tool that helped millions save automatically.

Variety Within the Design Thinking Discipline

1. Different Process Models

While the core idea remains similar (human-centered + iterative), organizations and educators adapt the steps:

Stanford d.school Model → Empathize → Define → Ideate → Prototype → Test

IDEO Model → Inspiration → Ideation → Implementation

Double Diamond Model (UK Design Council) → Discover → Define → Develop → Deliver

IBM Enterprise Design Thinking → Hills → Playbacks → Sponsor Users

Formation of a Design Thinking Team

Design thinking thrives on **diverse perspectives**, so team formation is deliberate:

1. Define the Goal

- Identify the problem or opportunity the team will address.
- Clarify the scope (product, service, policy, etc.).

2. Build a Multidisciplinary Team

Include members with varied expertise, such as:

- **Designers** – for creative ideation and visualization.
- **Engineers/Technologists** – for technical feasibility.
- **Business/Marketing Experts** – for market fit and strategy.
- **Researchers/Anthropologists** – for user insights.
- **End-Users/Stakeholders** – to provide real-world perspectives.

3. Assign Roles

- **Facilitator** – guides the design thinking process.
- **Research Lead** – manages user research and empathy work.
- **Idea Generator(s)** – pushes creative boundaries.
- **Prototyper(s)** – quickly turn ideas into tangible forms.
- **Tester(s)** – gather user feedback and evaluate solutions.

4. Set Team Norms

- Encourage open-mindedness.
- Accept failure as learning.
- Practice active listening.
- Promote equal voice in discussions.

Importance of a Design Thinking Team

1. Diversity Fuels Innovation

Different disciplines and viewpoints lead to more original, well-rounded solutions.

2. Human-Centered Outcomes

Teams work closely with users, ensuring solutions address real needs—not assumptions.

3. Faster Problem-Solving

Collaborative, iterative work cycles accelerate from idea to prototype to market.

4. Better Risk Management

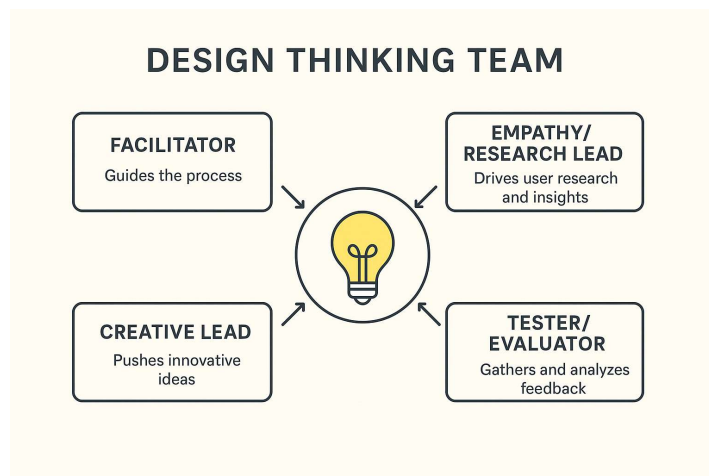
Testing and refining ideas early reduces costly failures later.

5. Organizational Impact

A strong design thinking team can foster a culture of innovation across the business.

💡 Key takeaway:

A design thinking team is like a **creative laboratory**—built on diverse skills, shared empathy, and rapid iteration. The right mix of people and roles can make the difference between an idea that fails quietly and one that transforms an industry.



Scenario: A school wants to improve the lunchtime experience for students

Design Thinking Team

Participants:

2 teachers

3 students from different grades

1 cafeteria manager

1 parent representative
1 design thinking facilitator

Scenario: A furniture manufacturing company wants to reduce assembly time for its modular office desks.

Design Thinking Team

Participants:

Production manager
2 assembly line workers
Quality control supervisor
Product designer
Supply chain manager
Design thinking facilitator

Scenario:

A software company is developing a mobile banking app and wants to improve the bill payment feature.

Design Thinking Team

Participants:

Product manager
2 UX/UI designers
1 backend developer
1 frontend developer
Customer support representative
Design thinking facilitator

Formation of Design Thinking Workshops & Meetings

Design thinking workshops and meetings are structured, collaborative sessions designed to help participants explore problems creatively and co-create solutions. Their formation usually follows these steps:

1. Define the Objective

Clearly state the purpose (e.g., understanding user needs, generating ideas, testing prototypes).

Link the workshop to a specific stage of the Design Thinking process (Empathize, Define, Ideate, Prototype, Test).

2. Select Participants

Include a diverse mix: designers, engineers, business experts, marketers, end-users, and decision-makers.

Diversity ensures different perspectives and prevents “groupthink.”

3. Prepare the Content & Tools

Gather research data, user personas, and any prior insights.

Arrange tools like sticky notes, markers, whiteboards, prototyping materials, or digital tools (e.g., Miro, FigJam).

4. Structure the Agenda

Keep activities time-boxed to maintain focus.

Example structure:

Warm-up (5–10 mins)

User insight review (15–20 mins)

Idea generation (30–60 mins)

Prototyping/sketching (60 mins)

Feedback & next steps (15–20 mins)

5. Assign Roles

Facilitator – keeps the session on track.

Scribe – documents ideas and decisions.

Timekeeper – ensures schedule is followed.

Participants actively contribute ideas and feedback.

6. Conduct & Capture Outcomes

Use visual thinking (drawings, diagrams, post-its) to make ideas tangible.

Document all outcomes for further analysis.

MUMBAI DABBAWALLAS THROUGH DESIGN THINKING



Case Study: Mumbai Dabbawallas through Design Thinking

Background

The Mumbai Dabbawallas are a world-renowned lunchbox delivery network in Mumbai. Every day, around 5,000 dabbawallas deliver over 200,000 tiffins with an accuracy rate of 99.99%. Their system has been studied by Harvard Business School, IIMs, and global business leaders as a benchmark for logistics, efficiency, and human-centered service design.

Applying the Design Thinking Stages

1. Empathize

- User Needs: Office-goers wanted fresh, home-cooked meals instead of canteen or restaurant food.
- Pain Point: Cooking at home is possible, but carrying tiffins through Mumbai's crowded trains and long commutes was stressful.
- Empathy Insight: The Dabbawallas understood that emotional connection to homemade food was central for the users (both employees and families).

2. Define

Problem Statement:

“How might we ensure that working professionals get fresh, home-cooked food on time every day despite Mumbai's chaotic traffic and long travel distances?”

3. Ideate

Ideas Explored:

- Use of trains as a backbone for long-distance delivery.
 - A color-coding and symbol system to overcome literacy barriers.
 - Low-cost, scalable, and manpower-driven logistics (vs. tech-heavy systems).
 - Community-based workforce ensuring trust and reliability.
-

4. Prototype

Developed a trial system:

- Collect tiffins from homes.
 - Transport via bicycles, handcarts, and trains.
 - Sort and re-route tiffins at railway stations using a unique coding system.
 - Deliver to offices before lunch hours.
 - Iterated over time to refine codes, timings, and team collaboration.
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5. Test

- The system was tested in real-life daily operations.
 - Outcomes:
 - Error rate of 1 in 16 million deliveries.
 - Adopted by lakhs of customers across Mumbai.
 - Created a livelihood for thousands of semi-literate workers.
-

Recognized by Forbes, Harvard, and the Six Sigma certification.

Design Thinking Principles in Action

Human-Centered: Focused on solving a daily-life problem of food accessibility.

Simplicity: Instead of expensive technology, used low-tech but ingenious coding.

Scalability: Expanded to serve over 200,000 tiffins daily.

Sustainability: Minimal cost, environment-friendly (use of trains, bicycles, and reusable tiffins).

Empowerment: Gave economic independence to thousands of dabbawallas.

Key Lessons for Design Thinking

1. Innovation doesn't always need high technology—simple, empathetic solutions can be transformative.
2. Understanding real human needs (emotional + practical) is more valuable than focusing only on efficiency.
3. Community-based systems can be as reliable as corporate logistics when designed thoughtfully.
4. Iteration and continuous refinement over the years made the system nearly error-free.

Link: [\(50\) Zomato can't beat this company run by Delivery Boys | Dabbawala Documentary - YouTube](#)

CASE STUDY: MUMBAI DABBAWALLAS THROUGH DESIGN THINKING

EMPATHIZE

User Needs

Office-goers wanted fresh, home-cooked meals instead of canteen or restaurant food



Pain Point

Cooking at home is possible, but carrying tiffins through Mumbai, crowded trains and long commutes was stressful

DEFINE

Ideas Explored

Use of trains as a backbone for long-distance delivery



Problem Statement

"How might we ensure that working professionals get fresh, home-cooked food on time every day despite Mumbai's chaotic traffic and long travel distances?"

IDEATE

Developed a trial system

- Collect tiffins from homes
- Transport via bicycles, handcarts, and trains
- Sort and re-route tiffins at railway stations using a unique coding system



Ideas Explored

- Use of trains as a backbone for long-distance delivery
- A color-coding and symbol system to overcome literacy barriers
- Low-cost, scalable, and manpower-driven logistics (vs, tech-heavy systems)

PROTOTYPE

Developed a trial system

- Collect tiffins from homes
- Transport via bicycles, handcarts, and trains
- Sort and re-route tiffins at railway stations using



Outcomes

The system was tested in real-life daily operations

- Error rate of 1 in 16 million deliveries
- Adopted by lakhs of customers across Mumbai

Case Study: Tata Nano through Design Thinking



Background

In 2008, Tata Motors launched the **Tata Nano**, known as the “world’s cheapest car” (₹1 lakh car). The idea was to provide safe and affordable mobility for millions of Indian families who were using two-wheelers. It was a **bold attempt at democratizing car ownership**, but despite its innovation, the car struggled in the market due to branding and perception challenges.

Applying Design Thinking Stages

1. Empathize

- **User Needs:** Millions of Indian families used **scooters/motorbikes** to carry 3–4 people, often in unsafe conditions.
- **Pain Points:**
 - Unsafe family travel on two-wheelers.
 - Aspiration to own a car, but affordability was the biggest barrier.
- **Empathy Insight:** The target audience desired **safety, dignity, and affordability** in mobility—not just a “cheap” car.

2. Define

- **Problem Statement:**

“How might we design a **safe, affordable, and reliable family car** for millions of Indian middle-class and lower-middle-class families currently dependent on two-wheelers?”

3. Ideate

Ideas Explored:

- A compact, fuel-efficient, low-cost car.
 - Minimalist design with essential features only.
 - Lightweight body with cost-saving innovations in manufacturing.
 - Focus on affordability without compromising **basic safety**.
-

4. Prototype

- Tata engineers created prototypes of a **compact, rear-engine car** with:
 - No unnecessary frills (like power steering, airbags initially)
 - Low-cost materials and simplified supply chains.
 - Innovative cost reductions in production and design (single wiper, fewer components, smaller size).
-

5. Test

- The Tata Nano was launched in 2008 as the “**people’s car**”.
- Initial excitement and global attention (praised as a frugal innovation).
- **Challenges in Testing & Market Feedback:**
- Branding as the “cheapest car” damaged aspirational value.
- Safety concerns (fire incidents hurt credibility).
- Middle-class families, the intended audience, avoided it due to image issues.
- Sales did not meet expectations; production was eventually discontinued in 2018.

Case Study: Tata Nano & Design Thinking



EMPATHIZE - UNDERSTANDING USERS

Millions of Indian families used 2-wheelers (scooters/motorbikes), often carrying 3-4 passengers. Safety was a huge issue—families were vulnerable to accidents in traffic and bad weather.



Tata Motors correctly identified the latent need for affordable mobility.



DEFINE - FRAMING THE PROBLEM

“How might we design a safe and affordable car for Indian middle- and lower-middle-class families who currently rely on two-wheelers?”



Focus on affordability + safety.



IDEATE - GENERATING POSSIBILITIES

Tata Motors explored ideas to minimize cost: Simplified design (no air conditioning, only one side mirror, single wiper), rear-engine layout to save space, innovative supply chain and vendor partnerships, modular manufacturing.



Radical cost-cutting ideas made car ownership accessible.



PROTOTYPE - MAKING IDEAS TANGIBLE

Prototypes of the Nano were created with bare minimum features to test feasibility.



Prototype proved the technical possibility of an ultra-low-cost car.



TEST - MARKET LAUNCH & FEEDBACK

Initial bookings were high (over 200,000), but sales quickly declined. Problems discovered during testing in

Case Study: Jeevan Bindi (Life Saving Dot)



Background

Problem: Millions of women in rural India suffer from iodine deficiency, which can cause goitre, impaired brain development, pregnancy complications, and fatigue.

Despite iodized salt being available, many rural families did not use it consistently due to a lack of awareness, affordability, or cultural food habits.

Women, however, wore bindis (traditional forehead dots) daily as part of their cultural identity.

1. Empathize

- Conducted field studies with tribal and rural women in Maharashtra.
- Discovered key barriers:
- Women did not take iodine supplements regularly.
- They did not recognize iodine deficiency as a serious health issue.

Insight: Women were more likely to adopt a solution if it integrated into their daily cultural practices.

2. Define

Problem Statement:

“How might we deliver a regular dose of iodine to rural women in a way that is simple, affordable, and culturally acceptable?”

3. Ideate

Explored creative solutions:

Fortified food packets? → Not consistently consumed.

Pills or drops? → Low compliance.

Wearable solution? → High adoption potential.

Best idea: Use the daily-worn bindi as a delivery system for iodine.

4. Prototype

Created Jeevan Bindi: an iodine-coated adhesive patch designed like a regular decorative bindi.

Each patch delivered 150 micrograms of iodine per day, meeting daily nutritional needs.

Prototypes were tested with rural women for comfort, wearability, and cultural acceptance.

5. Test

Piloted in villages of Maharashtra by Neelvasant Medical Foundation & Research Center with Grey for Good (Grey Group's philanthropic arm).

Results:

- Women accepted the bindis willingly (as it matched existing habits).
- Measured improvement in iodine levels among users.
- No side effects reported.
- Impact
- Health: Helped fight iodine deficiency in rural India.
- Accessibility: Extremely low-cost, easy to distribute, and culturally integrated.
- Awareness: Generated global recognition as a frugal innovation.

Featured in Time Magazine, The Better India, and showcased at multiple health innovation forums.

Why It's a Design Thinking Success

- ✓ Human-Centered: Designed around cultural practices, not against them.
- ✓ Frugal Innovation: Low-cost, high-impact solution for a widespread health issue.
- ✓ Simplicity: Turned a beauty accessory into a life-saving medical device.
- ✓ Scalability: Potential for use in other countries facing iodine deficiency.

[\(77\) Cannes Lions 015 - Life Saving Dot - YouTube](#)

CASE STUDY: JEEVAN BINDI & IODINE DEFICIENCY

1 BACKGROUND: THE DEFICIENCY PROBLEM



- Goiter
- Cognitive impairment
- Pregnancy complications
- Fatigue



Over 2 billion people



How might we provide women in rural India with a consistent source of iodine in a way that is low-cost, culturally acceptable, and easy to adopt?

5 PROTOTYPE & DEVELOPMENT



- Small adhesive dot
- Each delivers 100 micrograms of iodine per day through skin absorption

7 TESTING & RESULTS

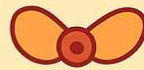


Piloted in rural Maharashtra by Neelvasant Medical Foundation & Grey for Good

2. EMPATIZE: UNDERSTANDING WOMEN IN RURAL INDIA



- Women in villages rarely consume iodine-rich foods
- Lack awareness of iodine's importance
- Consistently wear bindis daily as part of tradition



4. IDEATION & INNOVATION

Medical + cultural insight
The bindi is a daily-worn accessory



Idea

Create a bindi infused with iodine
→ Jeevan Bindi (Life Saving Dot)

6 IMPACT

- Health
- Cultural fit
- Cost-effective
- Global recognition



The Jeevan Bindi turns a symbol of beauty into a tool of health, transforming the fight against